

M00 / PROTECTED ANSWERS

M00 M01 Explained Reviews

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Read this first

Open this book after saving your own attempt. Compare the reasoning and result, repair the cause of a mistake, then close the answer and try changed inputs.

YOUR STANDALONE STUDY BOOK

Open the matching review after saving your attempt.

These explanations help you identify and repair mistakes. Keep the first attempt, compare its reasoning and actual output, then solve the changed retry independently. SQL answer files are divided into blocks so you can execute the matching example deliberately. Some temporary-object exercises depend on earlier blocks within that lesson. The learner book supplies that order.

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DE00-01 | Files, folders, drives and paths

First practice: explained review

Your path should end with `DE_Study\practice\first_attempt\location.txt`, although the part before `DE_Study` depends on your computer. The reopened document should still contain your recorded folder path. The two `location.txt` names have different parents, so Windows can store both. Seeing the name in Notepad's title is useful, but reopening it from the intended folder gives stronger evidence of a correct save.

Changed retry: explained review

Both folders should appear when you open `practice`. The first document keeps its original path text; the second contains "second attempt". If both documents changed, you probably reopened the same file twice. Compare the parents in File Explorer and retry the save with a distinct destination.

DE00-02 | File extensions and common data files

First practice: explained review

The file has three lines: one header and two records. Its final extension is `.csv`. It stores characters and separators, not an Excel chart object. Changing the filename alone leaves the same underlying CSV contents, so `.xlsx` would be a misleading label. An application must actually save or export a workbook format to create one.

Changed retry: explained review

The copy's name changes; the two lines inside it do not. The original `names.txt` still exists. Its `.txt` format stays the same. You should have two separate files, each containing Asha and Ravi.

DE00-03 | Download, ZIP, extract, copy, move and rename

First practice: explained review

The source `orders.csv` should still have `Pen,3`. The copy `orders-attempt-01.csv` should have `Pen,4`. Renaming affects only the selected item. If both now contain 4, you edited the source or opened the same path twice. Recover a fresh source from `file-practice.zip` and repeat using visible paths.

Changed retry: explained review

Recovery should contain Pen,3 because the archive holds the original input. The working copy's edit did not change the archive. This is the first version of an engineering habit: keep enough source material to reproduce or repair your work.

DE00-04 | Install only the software this stage needs

First practice: explained review

A download proves that the installer reached your device. A successful launch proves that an application can start in your account. Your setup record should distinguish these. "Installed" is not a claim you can make solely from seeing an EXE in Downloads. An honest blocked status is more useful than a false completion tick.

Changed retry: explained review

A successful relaunch confirms that the installed application remains available. Repeatedly running the installer is unnecessary. If blocked, the lesson remains pending practical setup; your diagnostic record is still useful evidence of what must be resolved.

DE00-05 | VS Code from zero

First practice: explained review

Both applications should display the same three saved lines. VS Code is an application for viewing and editing; a folder is a location containing files. If opening the evidence folder shows no file, inspect where Save As placed it. A screenshot of an unsaved tab alone does not prove that you created the required file.

Changed retry: explained review

There should be two files. Editing the copy should not change the original. If the original changed, you probably edited its tab. Verify the filename shown on the tab and reopen both through File Explorer.

DE00-06 | Terminal fundamentals

First practice: explained review

Use Set-Location -LiteralPath '.\evidence', Get-ChildItem, and Set-Location -LiteralPath '..'; Get-Location provides the final check. An evidence filename should match a real saved file from earlier lessons. Relative paths are interpreted from the shell's current location, so start by checking it when results surprise you.

Changed retry: explained review

The first navigation can use '.\practice\first_attempt'. Returning with the full path to evidence does not depend on where you currently are. Your final Get-Location should end with DE_Study\evidence. Explain this difference before you mark the practical check complete.

DE00-07 | Read errors and repair one cause at a time

First practice: explained review

The intended folder is practice; the attempt omitted the i. The evidence supports correcting that argument before changing software. Your ending path should end with DE_Study\practice. The repair record should preserve the failed attempt, the spelling difference and the check after correction.

Changed retry: explained review

Here the spelling is correct; the starting location is wrong for that relative path. Use '..\practice' from evidence, or the actual absolute path. Get-Location should confirm DE_Study\practice. This is a different cause from the first error, which is why memorising "fix spelling" is not enough.

DE00-08 | Study, evidence, attempts and review discipline

First practice: explained review

A path identifies a location; it does not automatically embed that file's contents in the study record. The backup therefore needs the study record and editor-attempt.txt. Reopening both copied files checks that the copies exist and contain the expected saved text. Your actual path may differ from another student's path. The important result is that you can locate and verify your own files.

Changed retry: explained review

The newer backup should contain the first attempt and retry. The earlier backup represents an earlier point in time; it does not update itself when you create another file elsewhere. A missing retry in that older copy is expected. A missing retry in the newer copy means you must inspect what you selected and where you copied it.

DE01-01 | What data is: tables, records, fields and data types

First practice: explained review

Structure, values and types

There are 4 records and 6 fields. One record describes one order containing one product kind. The header names fields; it does not describe an order. Order 0012 has quantity 4 and unit price 12.50: four pens, each priced at 12.50.

Types: text for order_id and item; integer for quantity; decimal amount for unit_price; date for ordered_on; Boolean with missing allowed for is_paid. Keep order_id as text because it identifies an order and its leading zeros belong to the original label.

Amounts with the reasoning visible

Order	Calculation	Amount	Category
0011	1 x 60.00	60.00	Known paid
0012	4 x 12.50	50.00	Recorded unpaid
0013	2 x 20.00	40.00	Unknown status
0014	1 x 250.00	250.00	Known paid

All orders: $60.00 + 50.00 + 40.00 + 250.00 = 400.00$. Known paid: $60.00 + 250.00 = 310.00$. Recorded unpaid: 50.00. Unknown status: 40.00.

Counts reconcile: $2 + 1 + 1 = 4$. Amounts reconcile: $310.00 + 50.00 + 40.00 = 400.00$. This supports consistency with the table and rules, not proof that every recorded payment matches reality.

Why the colleague's argument fails

Changing the missing status to FALSE changes the meaning of order 0013. Unpaid would become 90.00 and unknown would become 0.00, although the total stays 400.00. That claims knowledge the source did not provide.

A good answer can be simple: "Missing means we do not know. FALSE means we know it is unpaid. An unchanged total does not make that change correct." Your own wording is fine if the distinction remains clear.

If an answer differed

Counted 5 records? Revisit the table explanation. Chose numeric order IDs? Revisit the type explanation. Total 342.50? You added unit prices instead of row amounts: revisit the worked amounts. Classified missing as unpaid? Use the error example's counterexample. Repair the reason, not just the number.

Changed retry: explained review

First check the question

The request includes orders with a known payment status. Both TRUE and FALSE are known answers. Missing is not a known answer. Selecting only paid orders would solve the wrong problem.

One record is one order containing one product kind. There are 3 records and 6 fields. The header is not a fourth order.

Order	Calculation	Amount	Status
A015	5 x 8.00	40.00	Known paid
A016	2 x 5.00	10.00	Recorded unpaid

Order	Calculation	Amount	Status
A017	1 x 300.00	300.00	Unknown

Include A015 and A016: 2 records, 50.00. Unknown is A017: 1 record, 300.00. The whole-table total is 350.00.

Counts reconcile as $2 + 1 = 3$. Amounts reconcile as $50.00 + 300.00 = 350.00$. A017 is a text label containing a letter and digit characters; it identifies an order, not a quantity.

Repair the interpretation, not just the arithmetic

If you returned 40.00, you included TRUE but forgot FALSE is also known. If you returned 350.00, you included the unknown-status order in the known group. If you called A017 unpaid, you guessed a status that was not supplied.

Before doing arithmetic, say the inclusion rule aloud: "Include both known answers. Exclude the missing answer." Then identify the records. Calculating after selecting makes the reasoning easier to inspect.

Explain it in your own words

Try: "I included A015 and A016 because their statuses are supplied. Their amounts are 40.00 and 10.00, totalling 50.00. I kept A017 separate because its status is unknown. Together the known and unknown groups account for all three orders and 350.00."

You do not need to memorise that paragraph. You should be able to explain the selected records, the calculation, the excluded record and the check using your own simple words.

DE01-02 | Grain: what one row represents

First practice: explained review

Input: one product line per order. Output: one order. There is 1 order, 2 lines and 6 items. Goods are $50.00 + 40.00 = 90.00$. Payable is $90.00 + 15.00 = 105.00$. Summing both fees gives 30.00 and a wrong payable of 120.00. The repeated ID is expected at line grain.

Changed retry: explained review

Goods: $200.00 + 16.00 + 20.00 = 236.00$. Payable: 246.00. Three rows can be a valid line-level result, but the requested order summary needs one row. Adding 30.00 of delivery would incorrectly produce 266.00.

DE01-03 | Keys, duplicates and missing values

First practice: explained review

There are 4 source rows and 2 different supplied IDs: A1 and A2. One key-conflict group is A2, containing two rows with different amounts. One row lacks a key. Do not report four confirmed distinct orders or arbitrarily retain the first A2. Preserve the delivery and seek the rule or corrected source needed to resolve it.

Changed retry: explained review

(A2,1) repeats and needs inspection. (A1,1) and (A1,2) are different legitimate line keys. FALSE is recorded unpaid; missing is unknown. The repeated order ID and the missing status are separate questions, so one check cannot resolve both.

DE01-04 | Data quality and useful validation

First practice: explained review

Count 3 matches the stated delivery size. A11 repeats, so uniqueness fails. The -1 quantity breaks this exercise's positive-quantity rule. Three unknown statuses should be reported but do not break a rule that permits unknown. Preserve the input and record the two failures; ask for clarification instead of deleting a row or changing a sign.

Changed retry: explained review

Structural checks can all pass while the source is unsuitable for the decision. A month-old snapshot may omit later sales or deliveries. Obtain a sufficiently current source or label the result with its actual date and limitation. Do not present it as today's stock.

DE01-05 | Meet Excel: workbook, worksheet, cells and formulas

First practice: explained review

B4 is Folder; C2 is 2; F5 is genuinely empty, representing unknown status. J3 should contain $=3*10$ and display 30. The header names the fields; it is not another purchase. If you used a different sheet, its same cell addresses may hold different data.

Changed retry: explained review

Independent!C3 is 4 and D3 is 12.50. $=4*12.5$ gives 50. The two J2 cells belong to different sheets and can store different results. You will next replace hard-coded numbers with references that follow source changes.

DE01-06 | Enter, save and structure tabular data safely

First practice: explained review

There are two data records plus the header. IDs retain four characters. Quantities and prices are numeric. The date represents 6 September 2026. The final paid cell is empty, not the text "missing" and not FALSE. Two-decimal display changes appearance; it does not by itself convert text to a number or restore a lost ID.

Changed retry: explained review

Text supports labels containing letters as well as digit characters. A022 is recorded unpaid; 0021 remains unknown. Three records now exist. A new row should not cause you to replace the unknown answer in an earlier record.

DE01-07 | Sort and filter without damaging records

First practice: explained review

Expected order: 0008 first with quantity 3, then 0007 and 0010 with quantity 2 in ID order, then 0009 with quantity 1. The filtered Pen row remains 0008, quantity 3, unit price 10.00, date 2026-09-01, FALSE. Clear restores visibility of all four records, not their original order.

Changed retry: explained review

0012 has quantity 4 and 0013 has quantity 2, so both remain visible. The quantity-1 orders are hidden. Clearing the filter returns all four Independent records. Filtering is a view change, not permission to remove excluded source data.

DE01-08 | Basic formulas and essential functions

First practice: explained review

Amounts: 60, 50, 40, 250; total 400; mean 100; count 4. Groups: Paid 310, Unpaid 50, Unknown 40. Difference: 0. Temporarily doubling the first quantity adds 60, making total 460. Restoring it returns 400. This is evidence that the formulas use input references, not only copied answer numbers.

Changed retry: explained review

A015 gives 40, A016 gives 10 and A017 gives 300. Known status includes A015 and A016, totalling 50. Unknown is 300; full total is 350. Paid alone would be 40 and would answer a narrower question. You can add the Paid and Unpaid subtotals to calculate Known.

DE01-09 | Relative and absolute references

First practice: explained review

At 5% the amounts are 57.00, 47.50, 38.00 and 237.50, totalling 380.00. At 15% the total is 340.00. I5 should use G5 and \$K\$2. A correct first row alone does not establish that the copying pattern is correct.

Changed retry: explained review

At 12% the amounts are 35.20, 8.80 and 264.00, totalling 308.00. At 0% they return to the original full total of 350.00. The row amount reference moves; the single rate cell stays fixed. The test changes a shared input and checks every affected result.

DE01-10 | Excel Tables and structured working habits

First practice: explained review

The original total is 400. The added amount is 10, giving 410 across five records. The named total should read the actual Table amount column. Inspect the new row's formula and Table membership, not only the displayed total.

Changed retry: explained review

The extra eraser adds 5.00, so the temporary total is 415.00. Restoring quantity 2 returns 410.00. This checks the new row's formula and the summary reference together. It does not prove unrelated source values are correct.

DE01-11 | Practical cleaning and exact lookups

First practice: explained review

Marker maps to Writing; Notebook maps to Paper; Ruler remains unmatched. FALSE requires exact matching. The dollar signs keep the mapping rectangle fixed as each source row changes. Trimming normalises the permitted spaces; it does not add Ruler to the mapping table.

Changed retry: explained review

Pen now has two conflicting category assignments. A lookup may return one, but the mapping source no longer defines one unambiguous answer. Preserve both rows and ask which category rule is authoritative. Delete the experimental copy after preserving the attempt only if you no longer need it; keep the original Lookup intact.

DE01-12 | PivotTables and Power Query awareness

First practice: explained review

Four inputs become three category rows. Paper is 100, Writing 50, Bags 250, total 400. After the temporary change, Paper is 105 and total 405. Restoring the source and refreshing returns 400. Correct group totals must be accompanied by a correct source range and summary function.

Changed retry: explained review

Paper count 2, Writing count 1, Bags count 1, total count 4. It measures records, not currency. The same source supports different valid summaries when their labels and questions are explicit.

DE01-13 | Chart choice and basic visual communication

First practice: explained review

Bags is highest at 250. Its difference from Paper is 150. The source total is 400. The chart does not show profit, future demand or whether the recorded orders match real sales. Labels should identify the category and amount unit; the bar scale should begin at zero.

Changed retry: explained review

The measure has changed from amount to count. Counts sum to 4, and Paper now has the largest count despite Bags having the largest amount. A chart's interpretation depends on its measure and unit, not just the categories.

DE01-14 | Reconciliation, sanity checks and source preservation

First practice: explained review

Total is 400; categories are 310, 50 and 40; counts are 2, 1 and 1. Dropping 0013 gives 3 records and 360, which fails source comparisons even if internal groups sum to 360. Misclassifying it gives unpaid 90 and unknown 0 while the total remains 400; source-to-status classification catches this.

Changed retry: explained review

Count and whole total still pass. 0011's 60 becomes unpaid; 0012's 50 becomes paid. Paid becomes 300, unpaid 60, unknown 40. The total remains 400. Comparing statuses by their order IDs with the preserved source detects the swap.

DE01-15 | A complete small workflow: inspect, calculate, validate, explain and save

First practice: explained review

Amounts are 110, 45, 300 and 45; total 500. Paid is 155 across 2 records; unpaid is 45 across 1; unknown is 300 across 1. Amounts reconcile $155 + 45 + 300 = 500$ and counts $2 + 1 + 1 = 4$. C003 remains unknown. Your formulas, preserved source and explanation should all describe the same four records.

Changed retry: explained review

Amounts are 36, 18 and 44, total 98. Known status includes D001 and D002: 2 records and 54. Unknown is D003: 1 record and 44. Counts reconcile $2 + 1 = 3$; amounts $54 + 44 = 98$. Reporting only paid 36 would fail the changed question despite correct multiplication.